Vectors I - HW1 (Introduction to Ratio Theorem)

Solutions

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Original practice questions. Total: 52 marks. Give exact answers unless otherwise stated.

Question 1 **[6 marks]**

The points A, B, C and D have coordinates $(1, 2, 0)$, $(4, -2, 0)$, $(-2, 5, 1)$ and $(1, 1, 1)$ respectively.

(a) Find $\overrightarrow{AB}$, $\overrightarrow{CD}$ and the distance AB. [3]

(b) Are $\overrightarrow{AB}$ and $\overrightarrow{CD}$ equal vectors? Are $\overrightarrow{OA}$ and $\overrightarrow{OC}$ equal vectors? Explain the difference between these two comparisons. [2]

(c) A student claims that A, B, C, D are collinear because $\overrightarrow{AB} = \overrightarrow{CD}$. Explain why this is false. [1]

Solution

(a) $\overrightarrow{AB} = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 0 \end{pmatrix}$ and $\overrightarrow{CD} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 0 \end{pmatrix}$.

$$AB = |\overrightarrow{AB}| = \sqrt{3^2 + (-4)^2} = 5.$$

(b) Yes: $\overrightarrow{AB}$ and $\overrightarrow{CD}$ have the same components, hence the same magnitude and direction. No: A and C are different points, so their position vectors have different components. Equal displacement vectors need not start at the same point; equal position vectors, relative to the same origin, identify the same point.

(c) Parallel segments at different locations need not be collinear. Here every point of the line through A and B has $z = 0$, whereas C and D have $z = 1$.

Question 2 **[5 marks]**

The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 4\mathbf{j}$.

(a) Write $\mathbf{a}$ and $\mathbf{b}$ as column vectors and find $\mathbf{a} + 2\mathbf{b}$. [2]

(b) Find $\mathbf{c}$ such that $2\mathbf{a} - 3\mathbf{b} + \mathbf{c} = \mathbf{0}$. [2]

(c) Points P and Q have position vectors $\mathbf{a} + 2\mathbf{b}$ and $\mathbf{c}$ respectively. Find $\overrightarrow{PQ}$. [1]

Solution

(a) $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}$, $\mathbf{a} + 2\mathbf{b} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + 2\begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ 3 \end{pmatrix}$.

(b) $\mathbf{c} = 3\mathbf{b} - 2\mathbf{a} = \begin{pmatrix} -3 \\ 12 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} = \begin{pmatrix} -7 \\ 14 \\ -6 \end{pmatrix}$.

(c) $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix} -7 \\ 14 \\ -6 \end{pmatrix} - \begin{pmatrix} 0 \\ 7 \\ 3 \end{pmatrix} = \begin{pmatrix} -7 \\ 7 \\ -9 \end{pmatrix}$.

Question 3 **[5 marks]**

The vector $\mathbf{u}=\begin{pmatrix} t \\ 3 \\ t+2 \end{pmatrix}$ has magnitude $\sqrt{19}$, where t is real.

(a) Find all possible values of t. [3]

(b) Find the corresponding unit vectors in the direction of $\mathbf{u}$. [1]

(c) Are the two possible vectors $\mathbf{u}$ parallel? Justify your answer without calculating any angle. [1]

Solution

(a)
$$t^2+3^2+(t+2)^2=19 \quad \Longrightarrow \quad 2t^2+4t-6=0.$$
Thus $(t-1)(t+3)=0$, so $\boxed{t=1 \text{ or } t=-3}$.

(b) The unit vectors are $\dfrac{1}{\sqrt{19}}\begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$ and $\dfrac{1}{\sqrt{19}}\begin{pmatrix} -3 \\ 3 \\ -1 \end{pmatrix}$ respectively.

(c) No. If one were a scalar multiple of the other, their equal non-zero second components would force the scalar to be 1, but their first components are different.

Question 4 **[6 marks]**

The points A and B have coordinates $(1,-2,2)$ and $(3,1,-4)$ respectively.

(a) Find the unit vector in the direction of $\overrightarrow{AB}$. [2]

(b) Find the vector of magnitude 21 in the direction opposite to $\overrightarrow{AB}$. [2]

(c) A point P is such that $\overrightarrow{AP}$ has magnitude 14 and the same direction as $\overrightarrow{AB}$. Find the coordinates of P and state the order of the three points along their common straight path. [2]

Solution

(a)
$$\overrightarrow{AB}=\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix}-\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}=\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}, \qquad |\overrightarrow{AB}|=\sqrt{4+9+36}=7.$$
The required unit vector is $\dfrac{1}{7}\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}$.

(b) The required vector is $-21\left(\dfrac{1}{7}\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}\right)=\boxed{\begin{pmatrix} -6 \\ -9 \\ 18 \end{pmatrix}}$.

(c)
$$\overrightarrow{AP}=14\left(\frac{1}{7}\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}\right)=\begin{pmatrix} 4 \\ 6 \\ -12 \end{pmatrix}, \qquad \overrightarrow{OP}=\overrightarrow{OA}+\overrightarrow{AP}=\begin{pmatrix} 5 \\ 4 \\ -10 \end{pmatrix}.$$
Thus $\boxed{P(5,4,-10)}$. Since $\overrightarrow{AP}=2\overrightarrow{AB}$, the order is A, B, P.

Question 5 **[6 marks]**

The vectors $\mathbf{p}=\begin{pmatrix}6\\k\\-3\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}2\\-4\\m\end{pmatrix}$ are parallel, where k and m are real constants.

(a) Find k and m. [3]

(b) State whether $\mathbf{p}$ and $\mathbf{q}$ have the same or opposite directions, and find $|\mathbf{p}|:|\mathbf{q}|$. [1]

(c) Find all vectors of magnitude $\sqrt{21}$ that are parallel to $\mathbf{p}$. [2]

Solution

(a) Since the vectors are parallel, $\mathbf{p}=h\mathbf{q}$ for some non-zero real h. Comparing first components gives $6=2h$, so $h=3$.

$$k=3(-4)=-12, \qquad -3=3m \implies \boxed{k=-12,\ m=-1}.$$

(b) Since $\mathbf{p}=3\mathbf{q}$, they have the same direction and $\boxed{|\mathbf{p}|:|\mathbf{q}|=3:1}$.

(c) $|\mathbf{q}|=\sqrt{2^2+(-4)^2+(-1)^2}=\sqrt{21}$. Both directions must be included, so all such vectors are

$$\boxed{\pm\begin{pmatrix}2\\-4\\-1\end{pmatrix}}.$$

Question 6 **[6 marks]**

The points A, B and C have coordinates $(1,2,-1)$, $(3,1,2)$ and $(7,-1,8)$ respectively.

(a) Show that A, B and C are collinear, and find $AB:BC$. [3]

(b) The point $R(u,-2,v)$ lies on the same straight path as A, B and C. Find u and v. [3]

Solution

(a)

$$\overrightarrow{AB}=\begin{pmatrix}2\\-1\\3\end{pmatrix}, \qquad \overrightarrow{BC}=\begin{pmatrix}4\\-2\\6\end{pmatrix}=2\overrightarrow{AB}.$$

Thus $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are parallel and have the common point B, so A, B, C are collinear. Taking magnitudes gives $\boxed{AB:BC=1:2}$.

(b) Collinearity gives $\overrightarrow{AR}=s\overrightarrow{AB}$ for some real s, so

$$\begin{pmatrix}u-1\\-4\\v+1\end{pmatrix}=s\begin{pmatrix}2\\-1\\3\end{pmatrix}.$$

The second components give $s=4$. Hence $u-1=8$ and $v+1=12$, giving $\boxed{u=9,\ v=11}$.

Question 7 **[8 marks]**

The points A and B have coordinates $(2, -1, 3)$ and $(-4, 5, 0)$ respectively. A point M divides AB internally in the ratio $AM : MB = 2 : 1$. A point N lies on AB produced beyond B and satisfies $AN : NB = 5 : 2$.

(a) Find the position vector of M. [2]

(b) Find the position vector of N. [3]

(c) State the order of the four points and find $MB : BN$. [3]

Solution

(a) By the ratio theorem,

$$\overrightarrow{OM} = \frac{\overrightarrow{OA} + 2\overrightarrow{OB}}{3} = \frac{1}{3}\left[\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + 2\begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix}\right] = \boxed{\begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}}.$$

(b) Since $AN = AB + BN$ and $AN : BN = 5 : 2$, $AB : BN = 3 : 2$. Hence $\overrightarrow{BN} = \frac{2}{3}\overrightarrow{AB}$, and

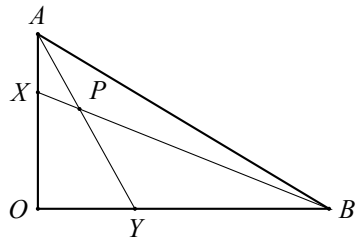
$$\overrightarrow{ON} = \overrightarrow{OB} + \frac{2}{3}(\overrightarrow{OB} - \overrightarrow{OA}) = \frac{5\overrightarrow{OB} - 2\overrightarrow{OA}}{3} = \boxed{\begin{pmatrix} -8 \\ 9 \\ -2 \end{pmatrix}}.$$

(c) The order is A, M, B, N. Since $MB = \frac{1}{3}AB$ and $BN = \frac{2}{3}AB$,

$$\boxed{MB : BN = 1 : 2}.$$

Question 8 **[10 marks]**

Relative to the origin O, the points A and B have non-zero, non-parallel position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point X lies on OA with $OX:XA=2:1$, and the point Y lies on OB with $OY:YB=1:2$. The line segments AY and BX intersect at P.



(a) By expressing $\overrightarrow{OP}$ in two ways using the ratio theorem, find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [6]

(b) Find $AP:PY$ and $BP:PX$. [2]

(c) The point Z divides AB internally with $AZ:ZB=1:4$. Show that O, P, Z are collinear and find $OP:PZ$. [2]

Solution

(a) First, $\overrightarrow{OX}=\frac{2}{3}\mathbf{a}$ and $\overrightarrow{OY}=\frac{1}{3}\mathbf{b}$. Let $AP:PY=s:(1-s)$ and $BP:PX=t:(1-t)$, where $0<s,t<1$. By the ratio theorem,

$$\overrightarrow{OP}=(1-s)\overrightarrow{OA}+s\overrightarrow{OY}=(1-s)\mathbf{a}+\frac{s}{3}\mathbf{b},$$
$$\overrightarrow{OP}=(1-t)\overrightarrow{OB}+t\overrightarrow{OX}=\frac{2t}{3}\mathbf{a}+(1-t)\mathbf{b}.$$

Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel, comparing coefficients gives

$$1-s=\frac{2t}{3}, \qquad \frac{s}{3}=1-t.$$

The second equation gives $s=3-3t$. Substituting into the first,

$$-2+3t=\frac{2t}{3} \quad\Rightarrow\quad 7t=6 \quad\Rightarrow\quad t=\frac{6}{7}, \quad s=\frac{3}{7}.$$

Consequently,

$$\boxed{\overrightarrow{OP}=\frac{4}{7}\mathbf{a}+\frac{1}{7}\mathbf{b}}.$$

(b) $\boxed{AP:PY=3:4, \qquad BP:PX=6:1}$.

(c) By the ratio theorem,

$$\overrightarrow{OZ}=\frac{4\mathbf{a}+\mathbf{b}}{5}, \qquad \overrightarrow{OP}=\frac{4\mathbf{a}+\mathbf{b}}{7}=\frac{5}{7}\overrightarrow{OZ}.$$

The two vectors are parallel and have the common point O, so O, P, Z are collinear. Since $0<\frac{5}{7}<1$, P is between O and Z; hence $\boxed{OP:PZ=5:2}$.

END OF HOMEWORK